

Finance Officer	Reviews financial information related to assets.	Views relevant costs, asset values, savings, and financial reports.	Reliable information about costs and financial savings.
Maintenance Staff	Inspects, maintains, and repairs assets.	Records inspections and repairs, updates maintenance status, and records costs.	Accesses asset condition and maintenance history.
Sustainability Officer	Monitors the environmental impact of resource management.	Views sustainability indicators and reports.	Monitors reuse, recycling, waste reduction, and environmental benefits.
System Administrator	Manages the system's technical operations and security.	Manages accounts, roles, permissions, and system configuration.	Maintains a secure and reliable system.

4. Requirements Gathering

4.1 Method

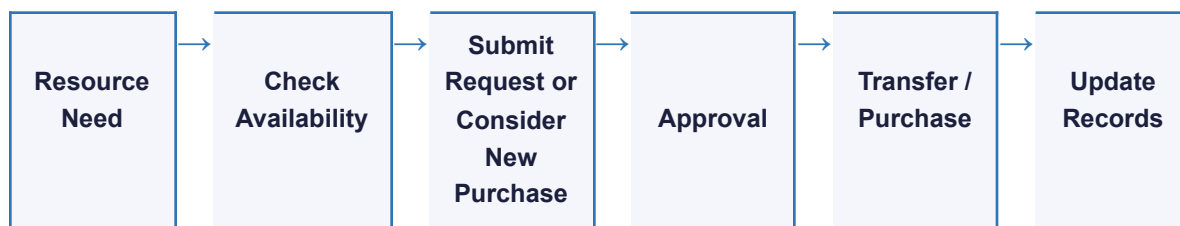
Method	Use in This Project
Document analysis	Use the official project description, Phase 1 slides, and Git/GitHub guide as the main source of required features and course rules.
Scenario / use-case review	Walk through tasks such as registering an asset, requesting an item, approving a transfer, and recording maintenance.
Instructor / stakeholder questions	Confirm unclear rules such as required fields, permissions, approval steps, and sustainability formulas.

Team review	Remove repeated requirements and check that each requirement is clear, useful, and realistic for the semester.
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4.2 Current Process Study

Currently, universities may purchase new resources while similar usable items remain unused in other departments. Traditional asset management systems mainly record information such as ownership, location, purchase information, and disposal status. This makes it difficult to discover reusable resources and match them with departmental needs.

Current Workflow:

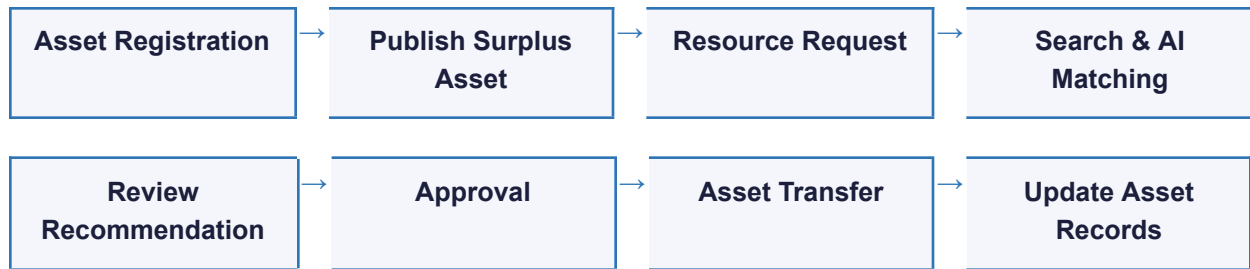


Limitations of the Current Process

- Departments may not easily know what resources are available in other departments.
- Usable resources may remain unused.
- Matching available assets with departmental requests is limited.
- New resources may be purchased even when reusable alternatives are available.
- Financial and environmental benefits may not be clearly measured.

4.3 Proposed Workflow

The proposed system will connect available campus resources with departmental needs through one integrated platform. Surplus assets can be registered and published, while other departments can search for resources or submit requests. The system can also use AI to recommend suitable matches based on factors such as technical compatibility, condition, quantity, location, and urgency.



The proposed workflow can reduce unnecessary purchases, improve resource reuse, and provide better tracking of assets throughout their life cycle. AI will support users with recommendations, while authorized users retain control of key approvals and decisions.

4.4 Priority of Requirements

Priority	Meaning / Examples
Must Have	Required core workflows: login/RBAC, assets, requests, search, approvals, transfers, maintenance, AI classification and matching, explanations, sustainability measures, and reports.
Should Have	Important features that may have less depth if time is short, such as richer dashboards, advanced notifications, and a more complete LLM assistant.
Could Have	Extra features after required work is stable, such as demand prediction, multilingual support, or advanced analytics.
Out of Scope	University-wide deployment, hardware tracking, and autonomous high-impact AI decisions.

5. Initial Requirements Specification

5.1 Classic Functional Requirements

ID	Requirement	Short Description
FR-01	User and role management	The system shall authenticate users and control access by role.
FR-02	Asset registration	Authorized users shall create and update asset records with basic details, condition, quantity, owner, location, value, photos/documents, and availability.
FR-03	Publish and search assets	Users shall publish available items and search/filter by useful fields such as category, condition, location, quantity, value, and status.
FR-04	Resource requests and reservations	Departments shall submit requests, reserve suitable items, and track request status.
FR-05	Approvals and transfers	The system shall support approval, collection/delivery, custody/location change, receipt, and transfer confirmation.
FR-06	Life-cycle and maintenance	The system shall record inspection, maintenance, repair, refurbishment, donation, recycling, retirement, and disposal events.
FR-07	Notifications and audit	The system shall notify users about important status changes and keep records of important actions and approvals.
FR-08	Operational reports	The system shall provide basic reports for assets, requests, transfers, maintenance, ownership, approvals, and status.

5.2 AI-Enhanced Functional Requirements

ID	Requirement	Short Description
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AI-01	AI asset classification	Suggest categories, tags, standard descriptions, material type, and missing information for a new asset.
AI-02	Semantic matching	Match requests with assets by meaning and important factors such as category, purpose, specifications, quantity, condition, location, urgency, date, transfer cost, and repair cost when available.
AI-03	Score, ranking, and explanation	Show a compatibility score, rank suitable options, and give a simple reason for each recommendation.
AI-04	Sustainable-action recommendation	Suggest reuse, transfer, repair, refurbishment, donation, recycling, or retirement, while keeping the final decision with an authorized user.
AI-05	LLM assistant	Allow natural-language questions for asset search, requests, history, status, policy guidance, and system navigation.
AI-06	AI-generated reporting	Create short summaries of exchange activity, maintenance trends, avoided purchases, savings, waste diversion, and estimated carbon reduction.

5.3 Sustainability and Quality Requirements

Area	Initial Requirement
Sustainability	Track reuse and waste diversion; estimate avoided purchases, savings, asset-life extension, and carbon reduction using stated assumptions.
Security / privacy	Use authentication, role-based access, secure storage, input validation, and controlled access to sensitive data.
Usability	Use clear and simple interfaces suitable for different campus roles.
Maintainability	Use modular design so major parts can be changed and tested separately.
Reliability / audit	Save important workflow changes correctly and keep useful audit records and error messages.

AI responsibility	Show explanations/confidence when useful, allow human review, and provide fallback behavior when AI is unavailable or uncertain.
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6. Preliminary Success Criteria

- All Must-Have workflows can be demonstrated in the final prototype.
- Users are allowed or blocked from actions according to their role.
- On a small test set, useful AI matches appear near the top when a suitable asset exists, and recommendations include a simple explanation.
- The required sustainability measures are calculated using documented assumptions and are clearly shown as estimates where needed.
- Testing results and system limits are reported clearly, and the main user flows are understandable during the class demonstration.

7. Initial Project Plan

- 7.1 Course Timeline and Deliverables

Phase	Timing	Deliverable	Weight
1	Week of Sep 14, 2026	Initial Plan and Requirement Gathering Document	10%
2	Week of Sep 21, 2026	Feasibility Document	10%
3	Week of Oct 5, 2026	Requirements Document	10%
4 & 5	Week of Oct 26, 2026	Design Document: Architecture and Detailed Design	20%
6	Week of Nov 16, 2026	Draft Implementation with Major Features	20%
7	Week of Nov 23, 2026	Details of Test Cases	10%
8	Week of Nov 23, 2026	Final Project, Presentation, and Demonstration	20%

7.2 Development Method

A simple hybrid method will be used. The course has fixed phase submissions, so major documents are completed in order. Inside each phase, the team will use short Agile-style tasks, regular reviews, small Git branches, early testing, and frequent integration. Must-Have work is completed before optional features.

7.3 Team and Resource Allocation

Role	Assigned Member	Main Responsibility
Project / Integration Lead	Maryam Alshehhi	Track schedule, tasks, dependencies, risks, and integration.
Requirements / Documentation Lead	Maryam Alshehhi	Maintain requirements, stakeholders, use cases, and phase documents.
Architecture / Data Lead	Ghala Alabdouli	Plan architecture, database, interfaces, and security boundaries.
Application / UI Lead	Ghala Alabdouli	Build core workflows, interface, business logic, and validation.
AI / Analytics Lead	Maitha Almaeeni	Build/integrate classification, matching, recommendations, assistant, and reports.
QA / Test Lead	Maitha Almaeeni	Prepare tests, track defects, and check required workflows.

7.4 Main Risks and Responses

Risk	Response
Scope becomes too large	Finish Must-Have work first and reduce optional depth.
Not enough realistic data	Use a documented sample/synthetic dataset and state its limits.
AI results are weak or AI service fails	Use explanations, test examples, human review, and a simple fallback.

Uneven team work or late integration	Assign owners, review Git activity, merge small changes often, and test combined workflows early.
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8. Version Control and Collaboration Plan

Git and GitHub will be used throughout the project. The team will create a shared GitHub repository and add the instructor and lab engineer as collaborators. Each member will use their own GitHub account and make regular, meaningful commits. Team members will work on separate branches and use pull requests when possible. Sensitive files such as passwords, API keys, and .env files will not be committed.

9. Open Items to Confirm

- The exact approval chain and permissions for release, transfer, donation, recycling, and disposal.
- The final required asset fields and whether real, anonymized, or synthetic data will be available.
- The sustainability formulas to use for savings, waste diversion, asset-life extension, and carbon estimates.
- The AI tools/services allowed by the course and how low-confidence results should be shown.

10. References

1. Khalifa University, COSC 336 Fall 2026, Intelligent, AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System - Project Description.
2. Khalifa University, COSC 336, Lab Phase 1 - Initial Plan and Requirement Gathering Document slides.
3. Git, GitHub, and Copilot for Senior Design - Version Control Onboarding guide.

Source note: The official course files define the project features and course rules. Details that depend on exact university operations are kept as assumptions or open questions until they are confirmed.